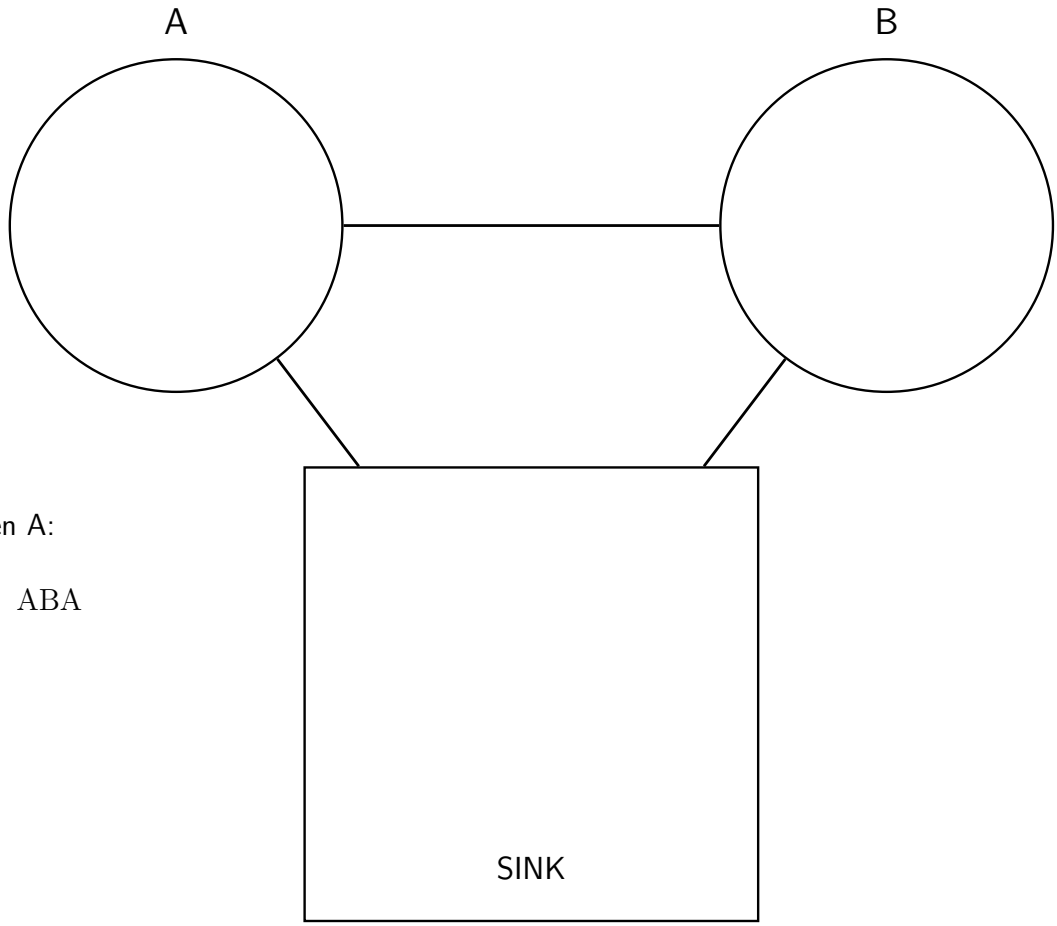
A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start. Write each sharing letter in order, then the final piles.

Problem 1: Try each starting pair and share until you must stop. Can choosing a different order leave different piles?

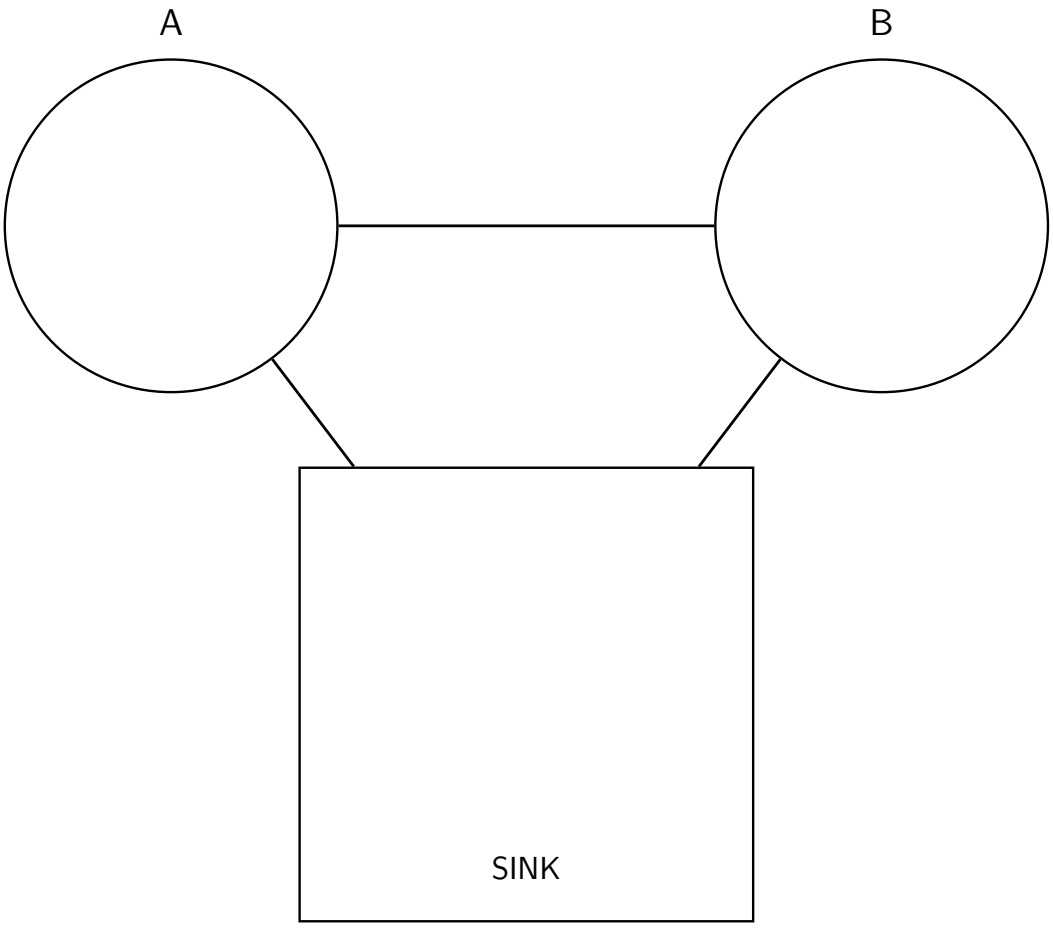


A, then B, then A:

$\mathrm{A} \rightarrow \mathrm{AB} \rightarrow \mathrm{ABA}$

Start A	Start B	Sharing word	Finish A	Finish B
● ●	● ●			
● ●	● ●			
● ● ●	● ●			
● ● ●	● ●			
● ●	● ● ●			
● ●	● ● ●			
● ● ●	● ● ●			
● ● ●	● ● ●			

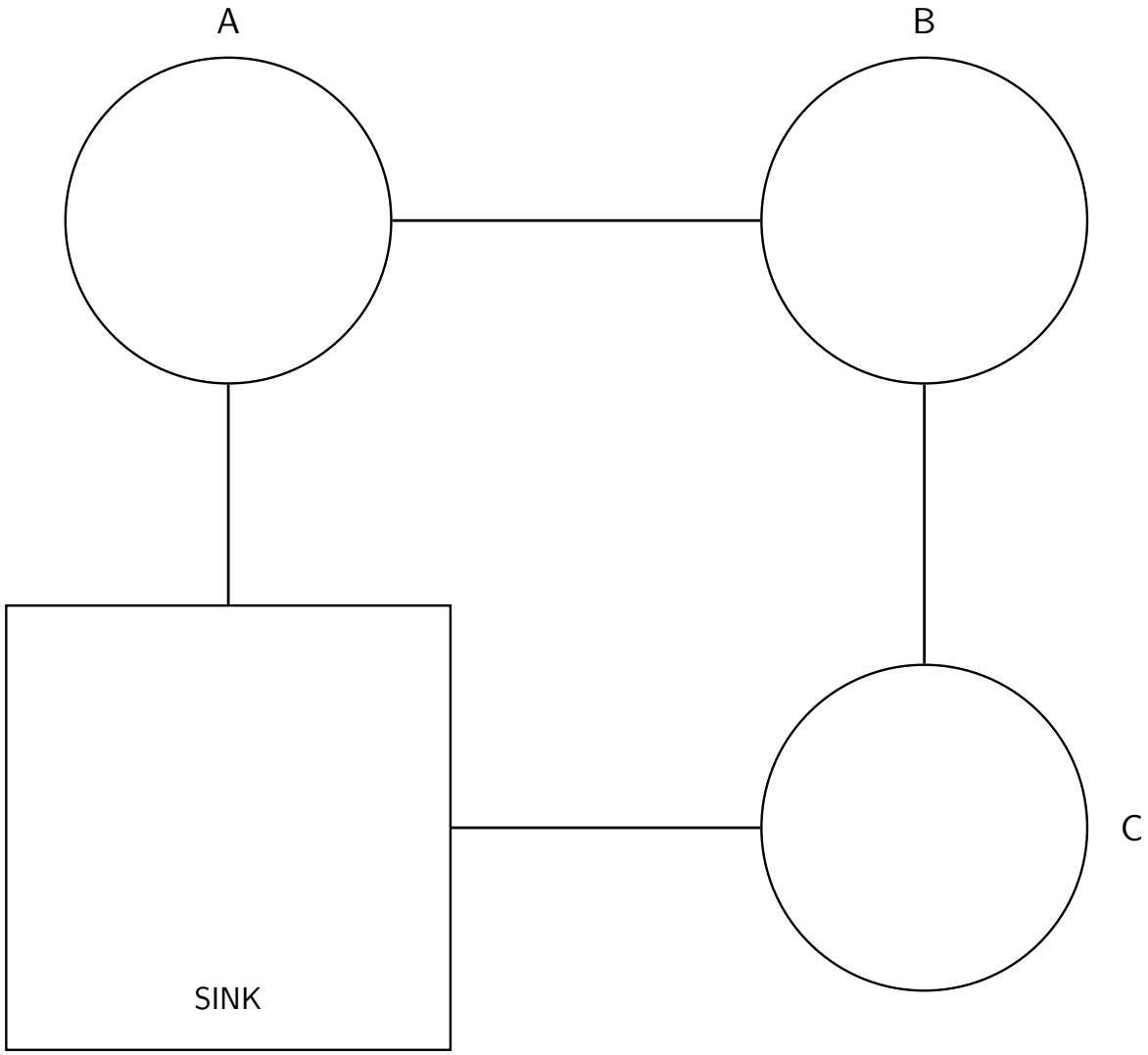
Problem 2: Put 4 chips altogether on A and B in any way, and share until you must stop. Find every finish; then try 5 chips.



4 chips

5 chips

Problem 3: Try each start and share until you must stop. Can choosing a different order leave different piles?



Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
0	● ● ● ●	0				
0	● ● ● ●	0				
● ●	● ●	0				
● ●	● ●	0				
0	● ●	● ●				
0	● ●	● ●				
● ●	0	● ●				
● ●	0	● ●				